Harmonic Shape Transform (HST) — Concept Document (2026)

General Scalar-Field–Based Framework for Shape Transformation

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1. Introduction

The Harmonic Shape Transform (HST) is a general mathematical framework for transforming one shape into another using normalized scalar fields defined on the shapes. Unlike classical correspondence-based methods, HST does not require matching topology, mesh connectivity, or explicit point-to-point alignment. Instead, it relies on the intrinsic structure of scalar fields to define smooth, stable, and topology-independent mappings.

The framework applies to 2D and 3D shapes, continuous domains, meshes, implicit surfaces, and volumetric objects. It is designed for applications in computer graphics, geometry processing, AI, generative modeling, biomechanics, and shape analysis.

2. Motivation

Traditional shape transformation methods face several limitations:

- They often require identical or similar topology.
- They depend on explicit correspondences, which are difficult to compute.
- They are sensitive to noise, discretization, and mesh quality.
- They do not generalize well across different shape categories.

HST addresses these issues by introducing a functional intermediary — a scalar field defined on each shape — and constructing the transformation through normalized level sets of these fields.

This enables:

- transformations between shapes with different topology,
- smooth and stable mappings,
- mathematically clean definitions,
- compatibility with AI embeddings and spectral geometry.

3. General Definition of the Harmonic Shape Transform (HST)

3.1 Scalar Field–Based Shape Transform

Let

$$\Omega_A, \Omega_B \subset \mathbb{R}^n$$

be two shapes (domains) equipped with smooth scalar fields

$$f_A: \Omega_A \to \mathbb{R}, f_B: \Omega_B \to \mathbb{R}.$$

These scalar fields serve as *functional intermediaries* between the shapes. They may represent, for example:

- Laplace eigenfunctions,

- heat kernel signatures (HKS),
- wave kernel signatures (WKS),
- signed distance functions (SDF),
- curvature-based functions,
- or any other smooth scalar descriptors.

To ensure comparability, we define normalized versions of these fields:

$$\tilde{f}_A = N_A(f_A), \tilde{f}_B = N_B(f_B),$$

where N_A, N_B are normalization operators (e.g., scaling to $[0, 1]$, standardization, or monotonic reparameterization).

3.2 Definition of the General HST

The **General Harmonic Shape Transform (HST)** from Ω_A to Ω_B induced by scalar fields f_A, f_B is any mapping

$$T_{f_A \to f_B} : \Omega_A \to \Omega_B$$

such that for each point $x \in \Omega_A$, the mapped point $y \in \Omega_B$ satisfies

$$\tilde{f}_B(y) \approx \tilde{f}_A(x).$$

Additional constraints may be applied, such as:

- spatial regularity,
- smoothness,
- energy minimization,
- probabilistic matching.

Intuitively, the transform aligns points on both shapes according to the normalized functional structure encoded by the scalar fields.

3.3 Special Case: Harmonic Note Transform

A key instance of HST arises when the scalar fields are Laplace eigenfunctions:

$$f_A = \phi_{A,k}, f_B = \phi_{B,k}.$$

After normalization, these become **harmonic notes**, and the resulting transform is the **Harmonic Note Transform**, a specific realization of the general HST framework.

This special case is:

- intrinsic,
- smooth,
- stable under deformation,
- independent of mesh connectivity,

- suitable for topology-independent mapping.

4. Example Use Case

Consider two shapes:

- **Shape A:** a smooth blob
- **Shape B:** a ring (different topology)

Steps:

1. Compute a scalar field on each shape (e.g., first Laplace eigenfunction).
2. Normalize both fields to $[0, 1]$.
3. Map each point on Shape A to a point on Shape B with the same normalized value.

This produces a smooth transformation even though the shapes have different topology.

5. Applications

Computer Graphics

- shape morphing
- animation
- procedural modeling
- mesh interpolation

AI and Generative Models

- functional embeddings for 3D data
- topology-independent latent spaces
- shape-to-shape generative models

Biomechanics and Medicine

- anatomical shape comparison
- organ and bone morphology
- statistical shape analysis

Geometry Processing

- shape matching
- correspondence-free registration
- spectral analysis

6. Advantages of HST

- Works across different topologies
- Smooth and stable

- Intrinsic and mathematically grounded
- Compatible with spectral geometry
- Generalizable to many scalar fields
- Suitable for AI integration

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